ViGoEmotions

Fine-grained emotion recognition
in Vietnamese social media

What changes when Neutral becomes a learned output?

Three models. Two label spaces. One shared 20-epoch protocol.

3
model families

20
epochs for every neural run

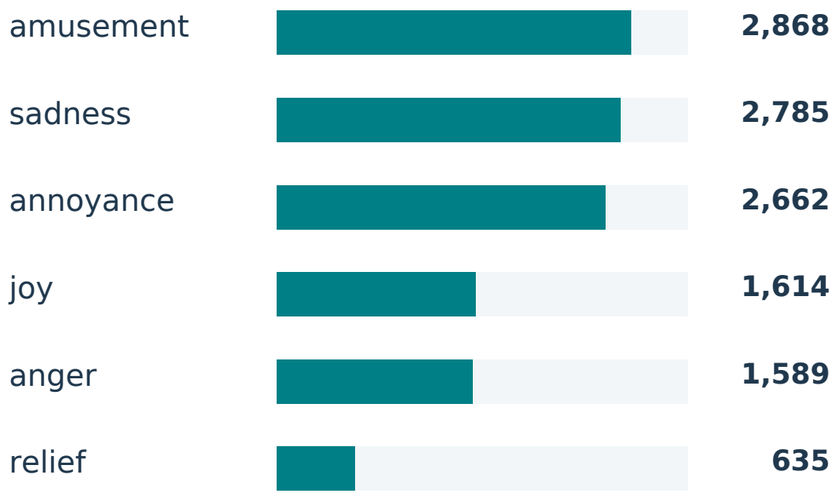
2
label settings: 27 and 28

Dataset, split and emotion frequency

20,664 source comments • fixed supplied partitions • label statistics from training data

Split	Source n	Used n	Role
Train	16,531	16,530	Fit model parameters
Validation	2,066	2,066	Select checkpoint / explore thresholds
Test	2,067	2,067	Held out from model evaluation
Total	20,664	20,663	One train–test text overlap removed

Most frequent five + least frequent label



1.859

emotions / train comment

11 / 47

median tokens / characters

TAKEAWAY **Preserve the split; fit text representations on train only.**

Sources: current study annotation_counts.csv; fixed split IDs; PhoBERT-27 per-label table. No test model scores.

27 vs 28 outputs: one additional decision

The label mapping and validation examples stay fixed; Neutral becomes an explicit target.

Property	27-emotion model	28-label model
Output space	27 emotion decisions	27 emotions + Neutral
Neutral-only target	All 27 emotion entries are zero	Neutral = 1; emotion entries = 0
Mixed annotation	Retain the emotion entries	Retain emotions and Neutral
Reported full Macro-F1	Average over 27 labels	Average over 28 labels

SAME INPUT, DIFFERENT TARGET SPACE

Target: joy + Neutral → **27 outputs: joy = 1** | **28 outputs: joy = 1, neutral = 1**

TAKEAWAY **A missing emotion prediction is not automatically a Neutral prediction.**

Sources: config27.json; evidence/28/config.json; annotation_counts.csv. Encoding examples are schematic targets.

Three routes from text to multi-label scores

Model-specific text representations • 27 or 28 independent label decisions

TF-IDF + LR

Lexical baseline

Regex tokens — 1–2 grams → **TF-IDF** — Fit vocabulary on train → **OvR classifiers** — One LR per label → **Label scores** — Threshold at 0.5

BIGRU

Learn from scratch

Token IDs — Lowercase + whitespace → **Embedding** — 128 dimensions → **BiGRU + linear** — Forward + backward → **Sigmoid scores** — 27 / 28 outputs

PHOBERT

Full fine-tuning

Segment + BPE — VnCoreNLP tokenizer → **PhoBERT backbone** — Pretrained Vietnamese → **Classification head** — Train with backbone → **Sigmoid scores** — 27 / 28 outputs

Shared preprocessing: Unicode NFC + whitespace normalization; retain Vietnamese accents and negation.

TAKEAWAY

Text representation changes across models; the label decision remains independent for each output.

Sources: frozen run configurations; train_neural.py; neutral_28_labels.py. Neural training: BCEWithLogitsLoss.

Architectures and training configurations

Shared seed 42 • fixed split • threshold 0.5 • Tesla T4 for neural models

Model	Representation / architecture	Parameters 27 / 28	Batch / learning rate	Budget
TF-IDF + OvR LR	Unigram/bigram; up to 40k features; liblinear, C = 1	0.941M / 0.975M	Not neural	max_iter = 1,000
BiGRU	Train from scratch; embed/hidden 128; length 128	2.128M / 2.129M	64 / 0.001	20 epochs
PhoBERT	VnCoreNLP + BPE; full fine-tuning; length 256	135.019M / 135.020M	16 / 2e-5	20 epochs

Select by validation emotion27 Macro-F1 at 0.5, including for the 28-output model; ties keep the earliest epoch.

TAKEAWAY **Equal epochs control the training budget; they do not imply equal compute.**

Sources: frozen manifests and six run.json files. Parameter counts shown as 27 / 28 outputs; rounded to millions.

PhoBERT leads within each label setting

Validation n = 2,066 • threshold 0.5 • separate full-label evaluations for 27 and 28 outputs

Labels	Model	Macro-F1 ↑	Micro-F1 ↑	Hamming ↓	LRAP ↑	Best ep.	Train time
27	TF-IDF + OvR LR	0.2388	0.2712	0.0606	0.6407	—	2.75 s
27	BiGRU	0.4220	0.4590	0.0626	0.6191	17	80.3 s
27	**PhoBERT**	**0.5974**	**0.5996**	**0.0550**	**0.7419**	14	4.05 h
28	TF-IDF + OvR LR	0.2310	0.2662	0.0601	0.6096	—	2.52 s
28	BiGRU	0.4086	0.4483	0.0628	0.5851	20	83.6 s
28	**PhoBERT**	**0.5783**	**0.5829**	**0.0575**	**0.7077**	15	3.90 h

Both neural models completed 20 epochs. “Best ep.” identifies the selected checkpoint, not the number of epochs trained.

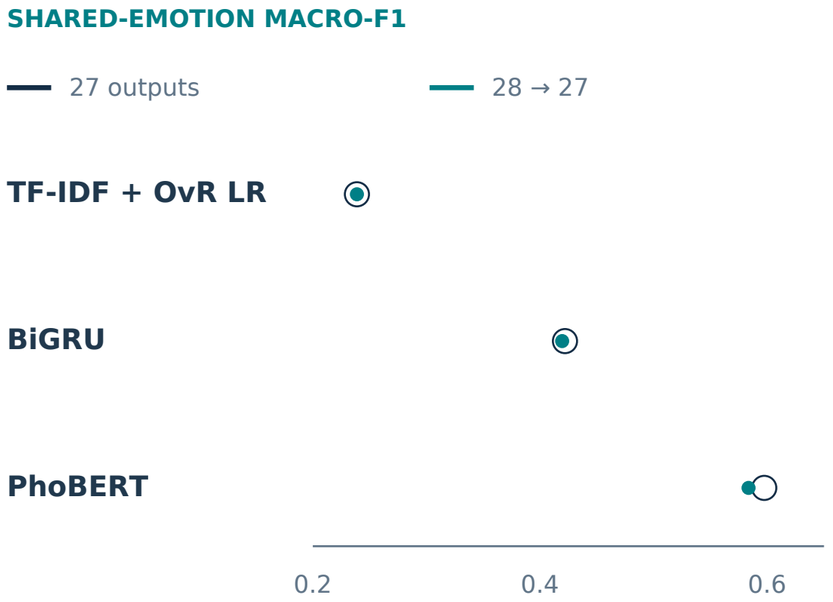
TAKEAWAY **Rank models within each block; the two Macro-F1 blocks evaluate different label sets.**

Source: model_comparison.csv. ↑ higher / ↓ lower is better. CPU TF-IDF; T4 neural. Times exclude preprocessing.

Adding Neutral does not raise shared-emotion F1 here

Same 27 emotion columns • same validation IDs • shared checkpoint selection • one seed

Model	27-output F1	28 → 27 F1	Δ (pp)
TF-IDF + OvR LR	0.2388	0.2388	+0.00
BiGRU	0.4220	0.4195	-0.25
PhoBERT	0.5974	0.5836	**-1.38**



This controls the epoch budget and scoring label space. It does not establish a general causal effect or statistical significance.

TAKEAWAY **PhoBERT changes by −1.38 percentage points on the shared 27-emotion evaluation.**

Source: paired_emotion27_delta.csv. Δ = 28-output projection minus 27-output score, in percentage points.

A learned Neutral output still makes substantial errors

28-label setting only • 94 Neutral-positive validation examples: 75 Neutral-only + 19 mixed

Model	Precision	Recall	F1	TP	FP	FN
TF-IDF + OvR LR	**0.5000**	0.0106	0.0208	1	1	93
BiGRU	0.1778	0.0851	0.1151	8	37	86
PhoBERT	0.4018	**0.4787**	**0.4369**	45	67	49

45
correct Neutral detections

49
missed Neutral examples

67
false Neutral predictions

An empty prediction is not a substitute for an explicitly trained Neutral label.

TAKEAWAY **PhoBERT finds 45 of 94 Neutral examples, with 67 false alarms.**

Source: neutral_metrics.csv. Threshold 0.5; support 94 for all models. No Neutral metric for the 27-output model.

Twenty epochs trained; the last checkpoint need not win

Both curves in each panel use the same 27-emotion checkpoint-selection metric.

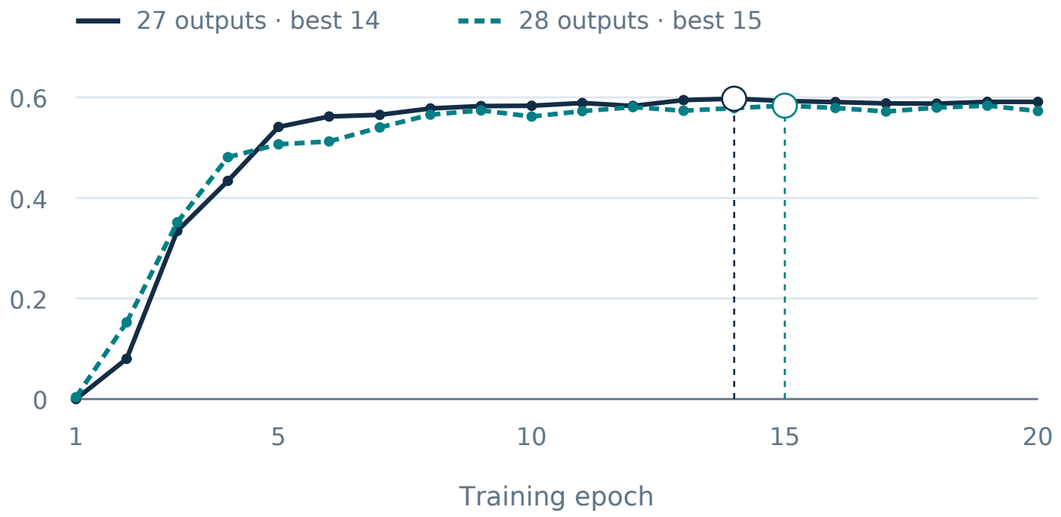
BiGRU

VALIDATION EMOTION27 MACRO-F1



PhoBERT

VALIDATION EMOTION27 MACRO-F1



Train BCE continued to decrease; validation loss was not recorded. These curves alone do not confirm overfitting.

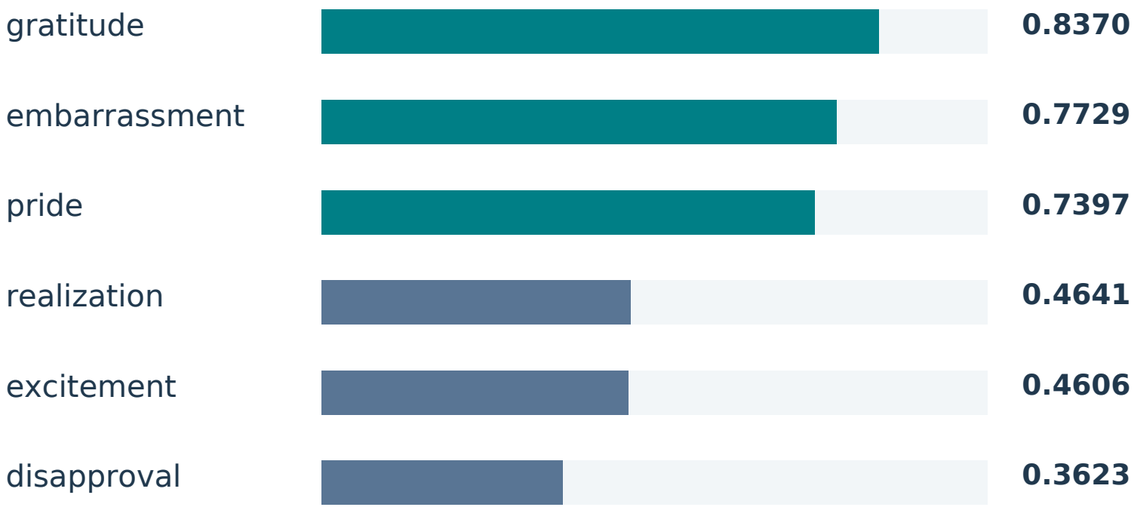
TAKEAWAY **Best PhoBERT checkpoints: epoch 14 for 27 outputs and epoch 15 for 28 outputs.**

Sources: evidence/{27,28}/{gru,phobert}/run.json. All histories contain epochs 1–20; no early stopping.

Q1 · Which emotions are easiest and hardest?

27-output study • per-label validation F1 at threshold 0.5 • bars show selected PhoBERT labels

PhoBERT: top three and bottom three labels



Model	Highest label / F1	Lowest label / F1
TF-IDF + OvR LR	gratitude 0.6797	realization 0.0000
BiGRU	gratitude 0.7701	realization 0.0940
PhoBERT	**gratitude 0.8370**	**disapproval 0.3623**

Per-label F1 reveals weaknesses hidden by the overall average. These are single-run validation results.

TAKEAWAY **Gratitude ranks highest for every model; the weakest label depends on the model.**

Sources: label_extremes.csv; current analysis/27_emotion27/*/tables/*_per_label.csv. Full label tables in report.

Q2 · Does label frequency predict F1?

27-output study • training counts versus validation per-label F1 • one point per emotion

PHOBERT VALIDATION PER-LABEL F1



Model	Spearman ρ	p-value
TF-IDF + OvR LR	0.242	0.223
BiGRU	0.313	0.112
PhoBERT	**0.018**	0.928

Frequency alone does not explain label difficulty. Association is not causation; one seed was evaluated.

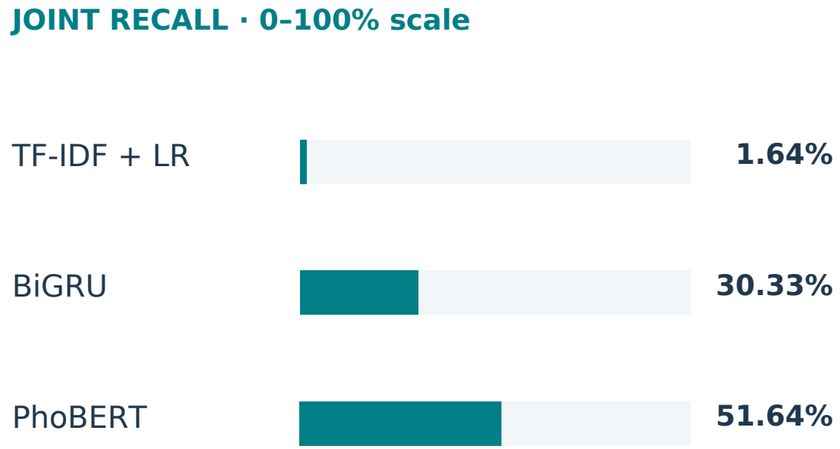
TAKEAWAY **PhoBERT shows almost no monotonic frequency–F1 association in this run.**

Sources: frequency_f1.csv; current PhoBERT-27 per-label table. Descriptive correlation across 27 labels per model.

Q3 · How often are both emotions recovered?

27-output study • threshold 0.5 • disappointment + sadness (top training pair: 904)

Model	Joint TP	Val. support	Joint recall
TF-IDF + OvR LR	2	122	1.64%
BiGRU	37	122	30.33%
PhoBERT	**63**	122	**51.64%**



Joint recall = cases with both labels predicted ÷ cases annotated with both labels.

The pair was selected by training co-occurrence, then measured on validation; extra labels do not invalidate a hit.

TAKEAWAY **PhoBERT recovers both emotions in 63/122 positive-pair cases; about half are still missed.**

Sources: pair_preservation.csv; current 27-output pair tables. Extra predicted labels allowed; this is joint recall, not precision.

Q4 · How does performance vary by text style?

27-output study • validation Macro-F1 over all 27 emotions • threshold 0.5

Text-style group	n	TF-IDF + LR	BiGRU	PhoBERT	Δ Pho−GRU (pp)
All comments	2,066	0.2388	0.4220	**0.5974**	+17.54
Emoji-heavy (≥ 2)	199	0.1690	0.3506	**0.4789**	+12.83
Short (≤ 5 tokens)	343	0.3310	0.4303	**0.5654**	+13.51
Teencode	307	0.1671	0.3333	**0.5547**	**+22.13**

Groups overlap and are heuristic subsets, not a partition. Short = ≤ 5 whitespace tokens; emoji-heavy = ≥ 2 emoji matches; teencode = lexicon match. Group sizes differ; no significance claim.

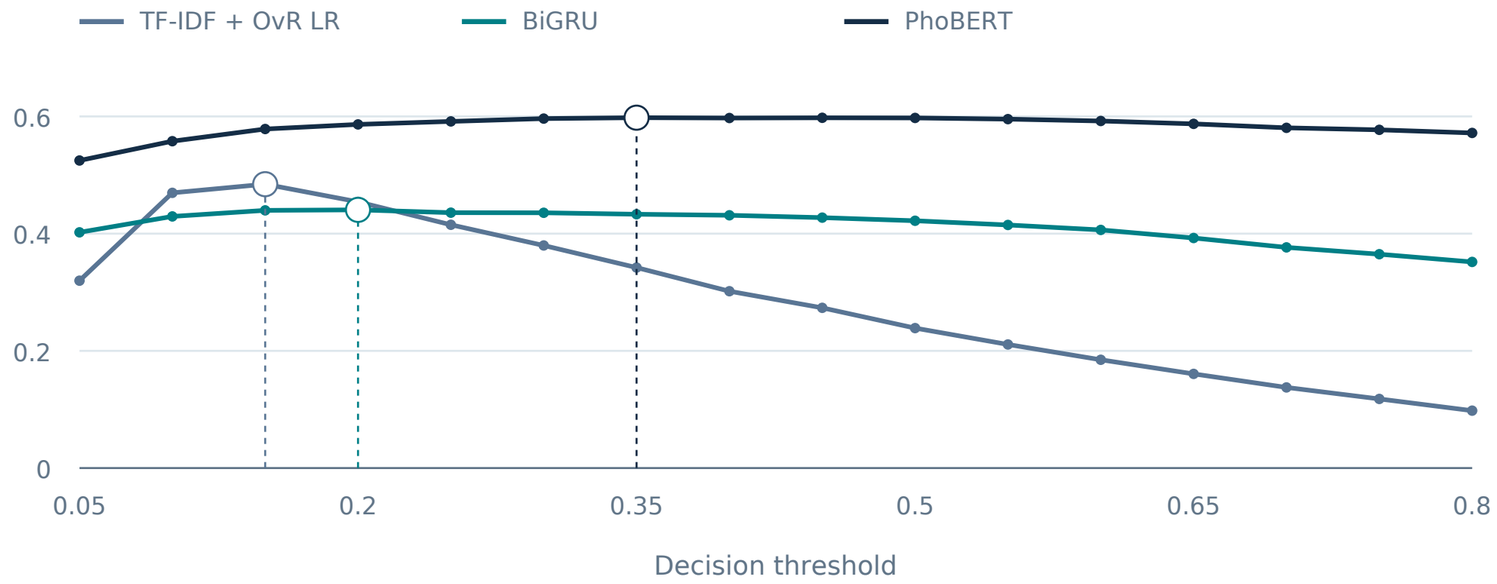
TAKEAWAY **PhoBERT leads in every style group; its largest gap over BiGRU is teencode (+22.13 pp).**

Sources: current analysis/27_emotion27/*/tables/*_styles.csv; neural_style_comparison.csv. Δ = PhoBERT − BiGRU.

Q5 · Lower thresholds affect models differently

27-output study • fixed checkpoint per model • validation grid: 0.05–0.80, step 0.05

VALIDATION MACRO-F1



+24.555 pp

TF-IDF + OvR LR · τ 0.50 → 0.15

+1.860 pp

BiGRU · τ 0.50 → 0.20

+0.043 pp

PhoBERT · τ 0.50 → 0.35

Callouts show the F1 change from 0.5 to the best grid point. Circles mark grid maxima; these are not test improvements.

TAKEAWAY **TF-IDF benefits most in exploratory F1; PhoBERT is almost unchanged. Keep the main comparison at 0.5.**

Source: current analysis/27_emotion27/*/tables/*_thresholds_val.csv. Grid 0.05–0.80, step 0.05.

Q5 · F1 gains come with more label errors

27-output study • same validation examples and checkpoint within each model • Macro-F1-selected grid points

Model	Threshold setting	τ	Macro-F1 ↑	Hamming ↓	Δ F1 (pp)
TF-IDF + OvR LR	Default	0.50	0.2388	0.0606	—
TF-IDF + OvR LR	Exploratory best	**0.15**	**0.4844**	0.0867	**+24.555**
BiGRU	Default	0.50	0.4220	0.0626	—
BiGRU	Exploratory best	**0.20**	**0.4406**	0.0784	**+1.860**
PhoBERT	Default	0.50	0.5974	0.0550	—
PhoBERT	Exploratory best	**0.35**	**0.5978**	0.0587	**+0.043**

Best = highest Macro-F1 on the explored validation grid, not an independently validated optimum. Main results remain at 0.5. LRAP is cutoff-independent; test generalization is unknown.

TAKEAWAY **Every best-F1 grid point has higher Hamming Loss; tuning F1 does not optimize every metric.**

Sources: model_comparison.csv (27 outputs); threshold_sensitivity.csv. Δ F1 versus each model’s default, in percentage points.

Neutral errors: implied affect versus surface cues

Two illustrative cases from the six-case preliminary, AI-assisted review; no raw comments reproduced.

FALSE POSITIVE / NEUTRAL

Validation ID 30099

Annotated: relief

Predicted: Neutral

P(Neutral) = 0.9965

Hypothesis: implied emotion in an explanatory comment.

FALSE NEGATIVE / NEUTRAL

Validation ID 6238

Annotated: Neutral

Predicted: anger + annoyance

P(Neutral) = 0.00005

Hypothesis: profanity may resemble negative affect.

Review scope: three confident Neutral false positives and three low-probability Neutral misses, selected deliberately.

TAKEAWAY

Interpretations are hypotheses from selected cases, not established error causes.

Source: phobert28_error_review.csv; current PhoBERT-28 validation predictions. Team confirmation remains pending.

Is one threshold suitable for every emotion?

Planned controlled decoding comparison • no new test results claimed

Freeze the best neural checkpoint: current PhoBERT-27

FIXED THRESHOLD

Use $\tau = 0.5$ for every emotion.

GLOBAL TUNED

Select one shared threshold on validation.

PER-LABEL TUNED

Select a separate threshold for each emotion.

1. Tune on validation

Compare decoders; review FP / FN

→

2. Lock all choices

Checkpoint + thresholds + protocol

→

3. Evaluate test once

Report scores and limitations

TAKEAWAY **Hold model weights fixed; change the decoder, then evaluate held-out test once.**

Source: course Final RQ1 and current midterm protocol. Thresholds selected strictly on validation predictions.

What the current study supports

Three evidence-backed observations; a focused next experiment

01 PhoBERT leads within both label spaces

Macro-F1: 0.5974 over 27 emotions; 0.5783 over all 28 labels.

02 Neutral is not a shared-emotion F1 gain here

The 28-output projection is 1.38 pp lower than PhoBERT-27.

03 Evaluate decoding with a frozen backbone

Explore fixed, global and per-label thresholds without retraining.

SCOPE AND LIMITATIONS

One seed; validation-based selection.

No model test scores or significance claims.

Original stratification algorithm unverified.

Heuristic styles; error review needs confirmation.

TAKEAWAY **PhoBERT is the strongest model in each setting; Neutral and decoder design remain open questions.**

Sources: current Kaggle 08–09 artifacts; ViGoEmotions (EACL 2026, doi:10.18653/v1/2026.eacl-long.129); PhoBERT (2020).